

WEB DESIGN

W3-S1 – CSS Concepts

HTML



HTML + CSS



CADT
IDT



Course Objectives



- ✓ Use CSS using either **inline**, **internal** or **external** syntax
- ✓ Manipulate **font and color properties** on HTML elements
- ✓ Use various **CSS selectors** to target HTML elements
 - **Type** Selector `h3 {}`
 - **ID** Selector `#id {}`
 - **Class** Selector `.class {}`
 - **Grouping** Selector `h3, .warning {}`
- ✓ Understand the **concept of CSS Inheritance**

What is CSS?

CSS is a language to describe the **appearance of an HTML document**.

HTML (STRUCTURE)

```
<body>  
  <h1>My First CSS Example</h1>  
</body>
```

CSS (APPEARANCE)

```
body {  
  background-color: lightblue;  
}  
  
h1 {  
  color: white;  
  text-align: center;  
}
```

My First CSS Example

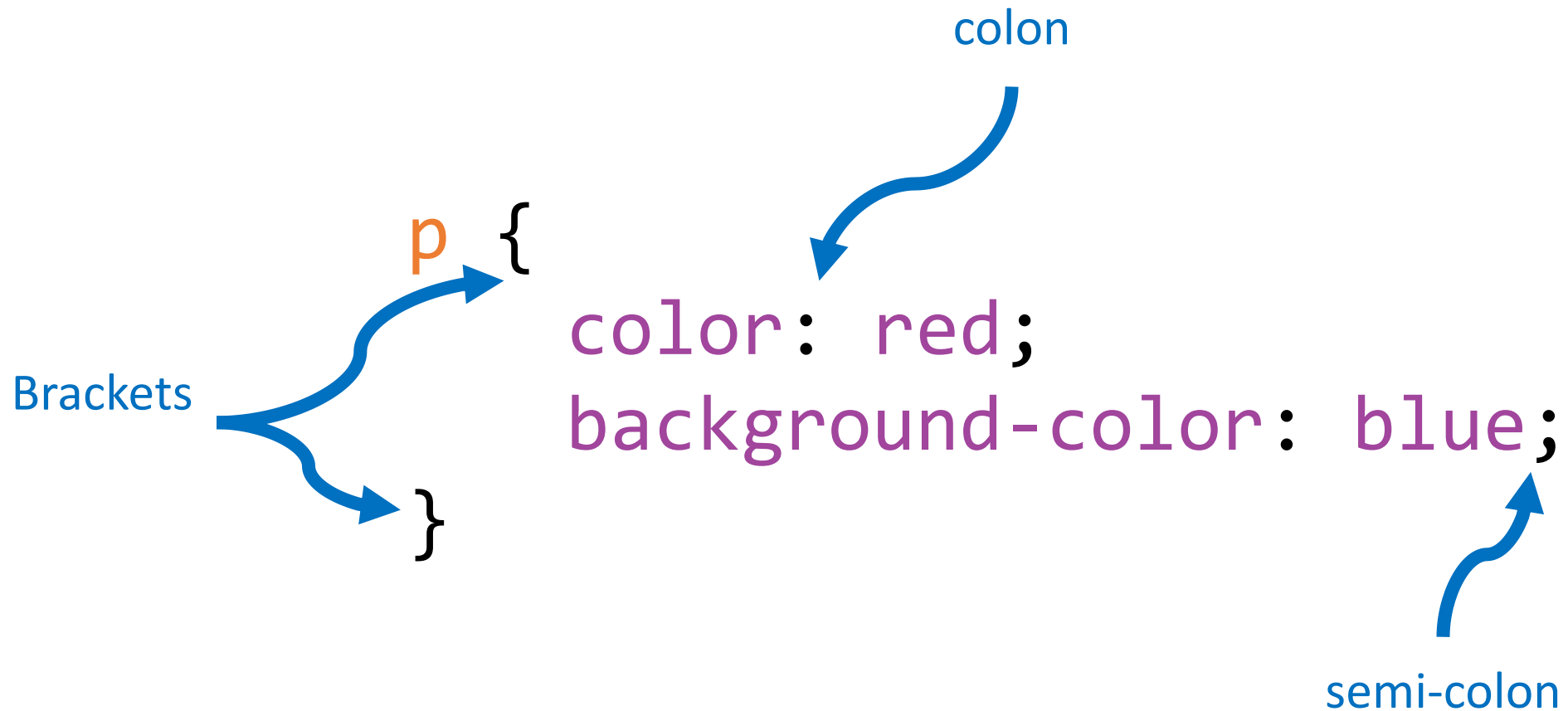
How to define a CSS rule?

A CSS **rule** is composed of a **selector** and a list of **definitions**



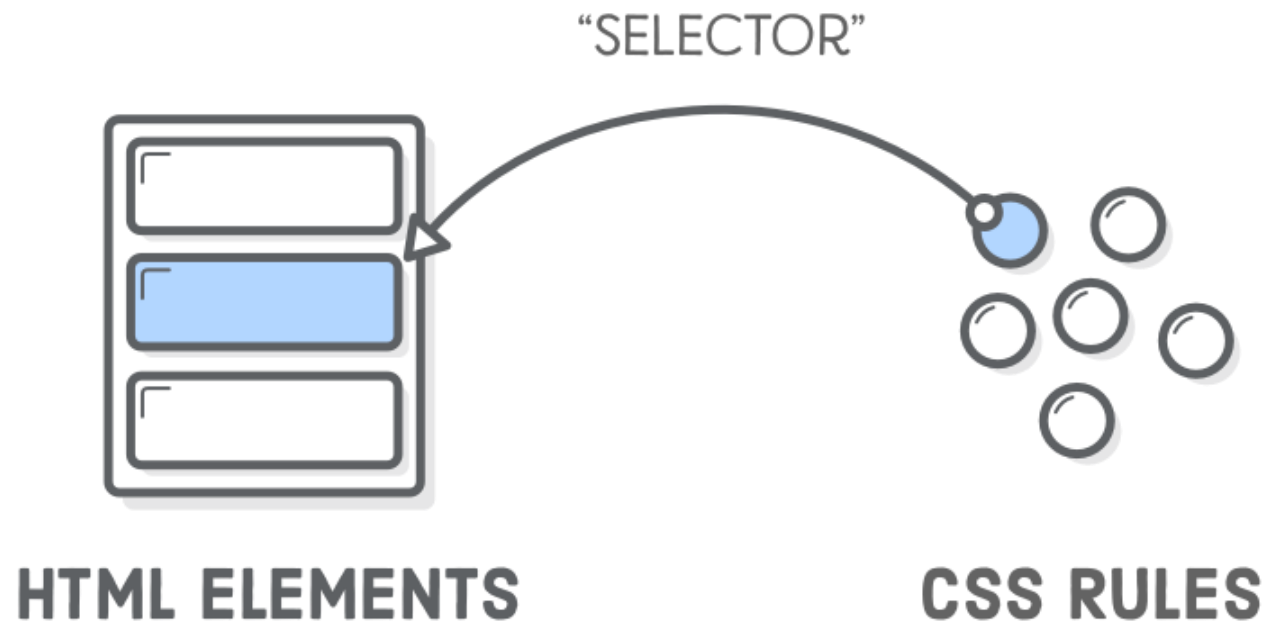
How to define a CSS rule?

Don't forget the required syntax



What is an CSS selector ?

A CSS selector **selects** the HTML element(s) you want to style



Types of CSS selector ?

Here are some type of selector we will learn during this course

- ✓ **Type** Selector `h3 {}`
- ✓ **ID** Selector `#id {}`
- ✓ **Class** Selector `.class {}`
- ✓ **Grouping** Selector `h3, .warning {}`
- ✓ **Child** Selector `ul > li {}`
- ✓ **Descendant** Selector `p a {}`
- ✓ **Universal** Selector `* {}`
- ✓ **Pseudo-classes** `p:hover {}`

Type Selector

A **type selector** selects all elements of the given type

Element Type



P

{

color:red;

}

How to link CSS to HTML?

CSS can be linked with the HTML using 3 ways

INLINE WAY

```
<h1 style="color:blue; font-size:14px;">Hello</h1>
```

BLOCK WAY

Use the tag **<style>** inside a HTML file

```
<head>  
  <style> h1 {color:blue; font-size:14px;} </style>  
</head>
```

EXTERNAL WAY

Define your rules in a separate file

```
<head>  
  <link rel="stylesheet" href="style.css">  
</head>>
```



That s the
usual way !

Identifier selectors

ON HTML SIDE

- ✓ An identifier can be assigned to an element using the Id attribute

ID attribute



```
<h1 id="main-title" >Welcome</h1>
```



ID

ON CSS SIDE

- ✓ The **ID selector** selects **the element** linked to this ID

ID selector



```
#main-title {  
    color:orange;  
}
```



An ID **should appear only once** in a HTML document.

Class selectors

ON HTML SIDE

- ✓ The **class attribute** is used to assign a class to an element

Class attribute
↓
<h1 class="title">Welcome</h1>
↑
Class name

ON CSS SIDE

- ✓ The **class selector** selects **all elements** linked to the class

Class selector
↓
.title {
 color:orange;
}

ID vs CLASSES

When should we use ID?, Classes?

```
#my-id {  
    color:orange;  
}
```

```
<p id="my-id" >Semester 1</p>  
<p>Semester 2</p>  
<p>Semester 3</p>  
<p>Semester 4</p>  
<p>Semester 5</p>
```

Semester 1
Semester 2
Semester 3
Semester 4
Semester 5

```
.warning {  
    color:orange;  
}
```

```
<p class="warning" >Semester 1</p>  
<p>Semester 2</p>  
<p>Semester 3</p>  
<p>Semester 4</p>  
<p class="warning" >Semester 5</p>
```

Semester 1
Semester 2
Semester 3
Semester 4
Semester 5

Div HTML Element

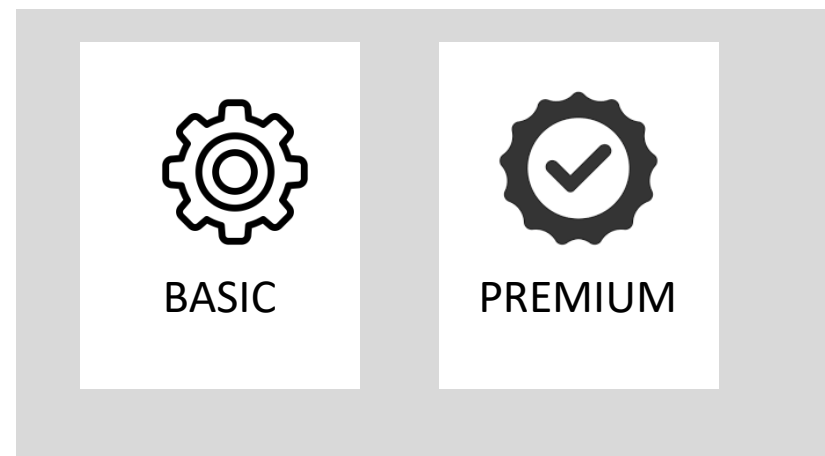
The `<div>` tag defines a division or a section in an HTML document.

- ✓ DIV is used as a **container** for HTML elements - which is then styled with CSS.

```
<div class="pricing-plan">
  <div class="pricing-item">
    
    <p>BASIC </p>
  </div>

  <div class="pricing-item">
    
    <p>PREMIUM</p>
  </div>
</div>
```

- . pricing-plan {background-color: grey;}
- . pricing-item {background-color: white;}



Cascade in CSS

- ✓ In CSS, the term "cascade" means that different styles can be used on the same HTML page
- ✓ These styles are applied to elements in a strictly defined order.

Cascading rules allow you to resolve situations where **multiple styles are specified for a single element**.



Cascade in CSS

What will be the **color** of P element?

```
#myP {  
  color: blue;  
}
```

```
.test {  
  color: green;  
}
```

```
p {  
  color: red;  
}
```

```
<body>  
  <p id="myP" class="test">Hello!</p>  
</body>
```

Cascade in CSS

P will be **blue** !

Highest priority

ID Selector

Class Selector

Element Selector

Lowest priority

This rule has the highest priority

```
#myP {  
  color: blue;  
}
```

```
.test {  
  color: green;  
}
```

```
p {  
  color: red;  
}
```

Grouping Selectors

The grouping selector selects all the HTML elements with the same style definitions

```
h1 {  
  text-align: center;  
  color: red;  
}
```

```
h2 {  
  text-align: center;  
  color: red;  
}
```

```
p {  
  text-align: center;  
  color: red;  
}
```



```
h1 , h2, p {  
  text-align: center;  
  color: red;  
}
```

- ✓ Easier to read
- ✓ Less duplication

We want to set all text color in green

```
<body>
```

```
  <h1>Title</h1>
```

```
  <h2>Heading 2</h2>
```

```
  <p>Paragraph
```

```
  <ul>
```

```
    <li>Item 1</li>
```

```
    <li>Item 2</li>
```

```
    <li>Item 3</li>
```

```
  </ul>
```

```
</body>
```

Title

Heading 2

Paragraph

- Item 1
- Item 2
- Item 3

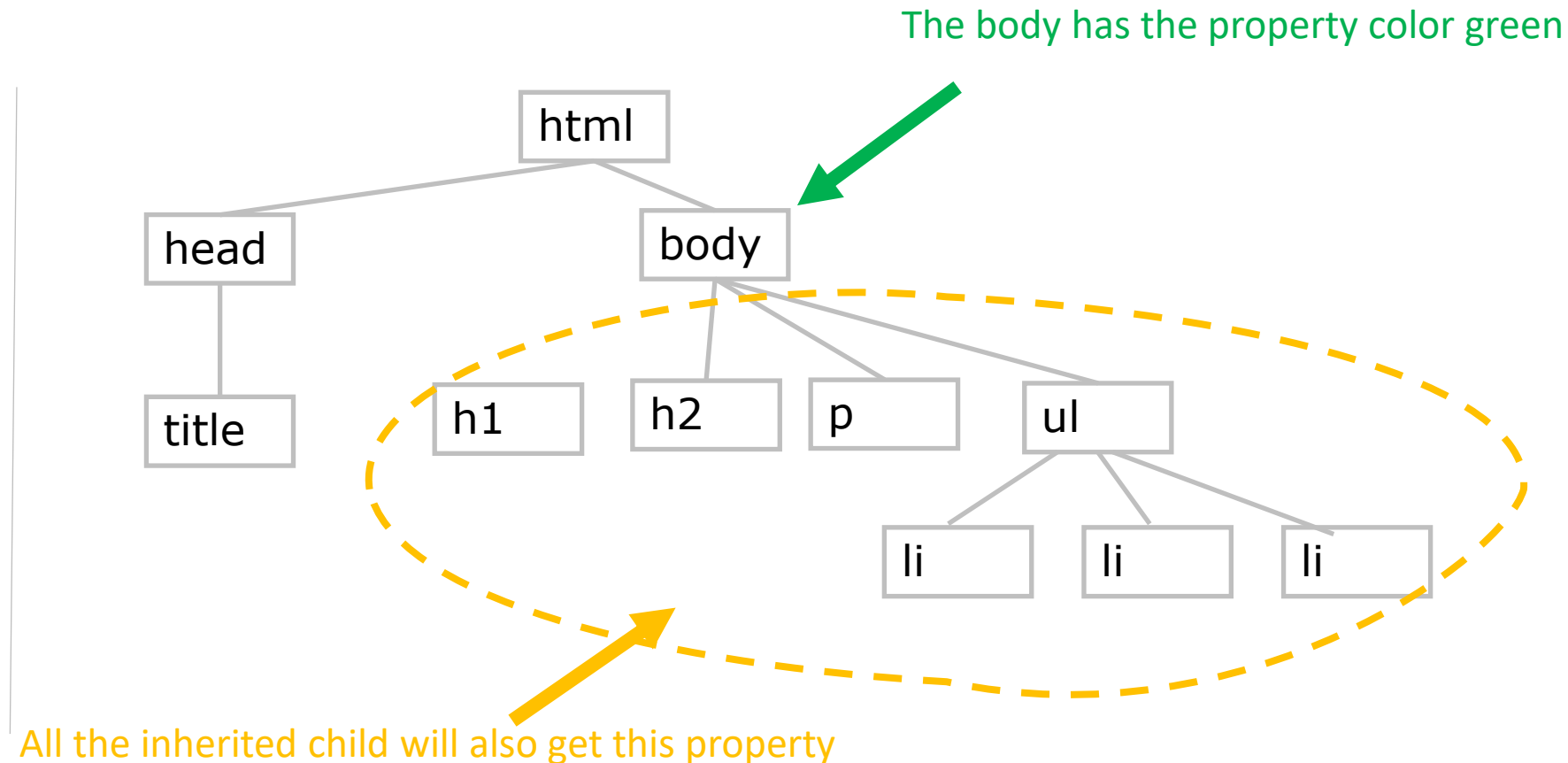
Can we find a **better way** than this code?

```
h1,h2,h3,p,a,strong,q {  
  color: green;  
}
```

Understanding CSS Inheritance

CSS inheritance is a mechanism that allows certain properties of a **parent element** to be automatically inherited by its **child elements**

```
body{  
  color: green;  
}
```





Congratulations !



You should now master those concepts

- ✓ Apply inline, internal, and external **CSS styles**
- ✓ Manipulate **font and color properties** on HTML elements
- ✓ Use various **CSS selectors** to target HTML elements
 - **Type** Selector `h3 {}`
 - **ID** Selector `#id {}`
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 - **Grouping** Selector `h3, .warning {}`
- ✓ Understand the **concept of CSS Inheritance**

3-2-1 Challenge

- ✓ List three things you **learned** today.
- ✓ List two **questions** you still have.
- ✓ List one aspect of the lesson or topic you **enjoyed**.



{ CSS Cheat Sheet }

<https://adam-marsden.co.uk/css-cheat-sheet>

Selectors

Universal Selector

`* {}`

ID Selector

`#id {}`

Class Selector

`.class {}`

Type Selector

`h1, h2, h3 {}`

Adjacent Sibling
Selector

`h1 + p {}`

Child Selector

`ul > li {}`

General Sibling
Selector

`h1 ~ p {}`

Descendant
Selector

`p a {}`

Attribute Selector

`div[attribute="SomeValue"] {}`

Pseudo Selectors & Elements

Mouse Over
Selector

`a:hover {}`

Active Link
Selector

`a:active {}`

Focus Selector

`input:focus {}`

A nice summary of everything you need to know on CSS

After this session !

PART 1 – LEARN MORE !

- ✓ Read more about CSS Selectors
<https://web.dev/learn/css/selectors>
- ✓ Read the CSS Cheat Sheet
<https://adam-marsden.co.uk/css-cheat-sheet>
- ✓ Another great article about CSS Selectors
<https://css-tricks.com/css-selectors/>

PART 2 – PLAY !

- ✓ Go as far as you can on this game
<https://flukeout.github.io/>

